**Example:** Determine

$$\int \frac{1}{x^2 - a^2} dx$$

for $a \neq 0$.

$\frac{1}{x^2-a^2} = \frac{1}{(x-a)(x+a)} = \frac{A}{x-a} + \frac{B}{x+a}$

Completing the square

How do we determine what A and B are?

cross multiply

Splitting a product into a sum

But to add fractions they need to have a common denominator

these are fancy 1's

$\frac{A}{x-a}\frac{(x+a)}{(x+a)} + \frac{B}{(x+a)}\frac{(x-a)}{(x-a)}$

$1 = A(x+a) + B(x-a)$

If $x = a$: $1 = 2Aa$, $A = \frac{1}{2a}$

If $x = -a$: $1 = B(-2a)$, $B = \frac{-1}{2a}$

$\int \frac{1}{x^2-a^2}dx = \frac{1}{2a}\int\frac{1}{x-a}dx - \frac{1}{2a}\int\frac{1}{x+a}dx$

$u = x-a$, $du = dx$; $u_2 = x+a$, $du_2 = dx$

$= \frac{1}{2a}\int\frac{1}{u}du - \frac{1}{2a}\int\frac{1}{u_2}du_2$

$= \frac{1}{2a}\ln|u| - \frac{1}{2a}\ln|u_2| + C$

$= \frac{1}{2a}\ln|x-a| - \frac{1}{2a}\ln|x+a| + C$

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**Partial Fraction Decomposition:** A rational function $f(x)$ takes the form

$$f(x) = \frac{p(x)}{q(x)},$$

where degree$(p(x))$ < degree$(q(x))$. To integrate, it is often easier to express a rational function as a sum of simpler rational functions:

$$f(x) = f_1(x) + f_2(x) + \cdots + f_k(x),$$

where each $f_i(x)$ takes the form

Linear $\frac{A}{(ax+b)^n}$ or quadratic $\frac{Ax+B}{(ax^2+bx+c)^n}$.

Given a rational function $f(x)$ of form $f(x) = \frac{p(x)}{q(x)}$ with polynomials $p(x)$ and $q(x)$ — degree of $q(x)$ to be bigger than degree of $p(x)$.
Decompose $f(x) = F_1(x) + F_2(x) + \dots F_k(x)$
where $F_i(x) = \frac{A}{(ax+b)^n}$ (Linear) or $F_i(x) = \frac{Ax+B}{(ax^2+bx+c)^n}$ (quadratic)

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**Example:** Find

$$\int \frac{x+2}{(x^2+2x+1)(x-1)} dx.$$

$\frac{x+2}{(x^2+2x+1)(x-1)} = \frac{x+2}{(x+1)^2(x-1)}$

$\frac{x+2}{(x+1)^2(x-1)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x-1}$

Every iteration - If it was $(x+1)^3$ you'd have $x+1$, $(x+1)^2$, and $(x+1)^3$. You want every power up to the one represented

To get common denominator: $x+2 = A(x+1)(x-1) + B(x-1) + C(x+1)^2$

was this the cross mult?

Now solve for A B and C. Choices need to hold for any val of x

If $x=1$: $3 = C4$, $C = \frac{3}{4}$

If $x = -1$, $1 = -2B$, $B = -\frac{1}{2}$

If $x = 0$: $2 = -A - B + C$

$2 = -A + \frac{1}{2} + \frac{3}{4}$

$A = -\frac{3}{4}$

once you solve for these they aren't functions of x and these are constants that will hold for any value x

$-\frac{3}{4}\int\frac{1}{x+1}dx - \frac{1}{2}\int\frac{1}{(x+1)^2}dx + \frac{3}{4}\int\frac{1}{x-1}dx$

$u = x+1$, $du = dx$; $u_2 = x-1$, $du_2 = dx$

$-\frac{3}{4}\int\frac{1}{u}du - \frac{1}{2}\int\frac{1}{u^2}du + \frac{3}{4}\int\frac{1}{u_2}du_2 = -\frac{3}{4}\ln|x+1| + \frac{1}{2}\frac{1}{x+1} + \frac{3}{4}\ln|x-1| + C$

Partial fraction decomposition - taking a rational expression and decomposing it into simpler rational expressions that you can add or subtract to get the original expression

$$f(x) = \frac{P(x)}{Q(x)}$$ where both $P(x)$ and $Q(x)$ are polynomials and degree of $P(x)$ is smaller than degree of $Q(x)$

Degree of a polynomial = largest exponent in the polynomial

Partial fractions can only be done if the degree of the numerator is strictly less than the degree of the denominator - important to remember!

| Factor in the denominator | Term in partial fraction decomposition |
|---|---|
| $ax+b$ | $\frac{A}{ax+b}$ |
| $(ax+b)^k$ | $\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \dots + \frac{A_k}{(ax+b)^k}$, $k = 1, 2, 3 \dots$ (counting up to value of $k$?) |
| $ax^2+bx+c$ | $\frac{Ax+B}{ax^2+bx+c}$ |
| $(ax^2+bx+c)^k$ | $\frac{A_1x+B_1}{ax^2+bx+c} + \frac{A_2x+B_2}{(ax^2+bx+c)^2} + \dots + \frac{A_kx+B_k}{(ax^2+bx+c)^k}$, $k = 1, 2, 3 \dots$ (counting up to value of $k$) |

When looking at $x^2$ in the denominator...

$$x^2 = (x-0)^2 \rightarrow \frac{A}{x} + \frac{B}{x^2}$$

What about $x^3$ or $x^4$?

$$x^3 = (x-0)^3 \rightarrow \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3}$$

$x^3$ is a repeated linear factor $(x \cdot x \cdot x)$ so we must include terms for each power

$$x^4 = (x-0)^4 \rightarrow \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{x^4}$$

Method of Partial Fractions when $\frac{f(x)}{g(x)}$ is proper ($\deg f(x) < \deg g(x)$)

1. Let $x-a$ be a linear factor of $g(x)$. Suppose that $(x-a)^k$ is the highest power of $x-a$ that divides $g(x)$. Then, to factor, assign the sum of the $k$ partial fractions:

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \frac{A_3}{(x-a)^3} + \cdots + \frac{A_k}{(x-a)^k}$$ Do this for each distinct linear factor of $g(x)$.

2. Let $x^2+px+q$ be an irreducible quadratic factor of $g(x)$ so that $x^2+px+q$ has no real roots. Suppose that $(x^2+px+q)^n$ is the highest power of this factor that divides $g(x)$. Then, to this factor, assign the sum of the $n$ partial fractions:

$$\frac{B_1x+C_1}{(x^2+px+q)} + \frac{B_2x+C_2}{(x^2+px+q)^2} + \frac{B_3x+C_3}{(x^2+px+q)^3} + \cdots + \frac{B_nx+C_n}{(x^2+px+q)^n}$$ Do this for each distinct quadratic factor of $g(x)$

3. Continue with this process with all irreducible factors and all powers. Things to remember are:
   (i) one fraction for each power of the irreducible factor that appears
   (ii) Degree of num should be one less denom

4. Set the original fraction $\frac{f(x)}{g(x)}$ equal to the sum of all these partial fractions. Clear the resulting equation of fractions and arrange the terms in the decreasing powers of $x$

5. Solve for the undetermined coefficients by either plugging in values or comparing coefficients of powers of $x$

What happens when degree($p(x)$) $\geq$ degree($q(x)$)?

**Example:** Find

$$\int \frac{x^3 - 6x^2 + 5x - 3}{x^2 - 1} dx.$$

Degree of top is 3 and bottom is 2 so this breaks the rule! So we need to get this in a workable form first

Polynomial long division

$$\begin{array}{r} x - 6 \\ x^2 + 0x - 1 \overline{\big) x^3 - 6x^2 + 5x - 3} \\ -(x^3 + 0x^2 - x) \\ \hline -6x^2 + 6x - 3 \\ -(-6x^2 + 0x + 6) \\ \hline 6x - 9 \end{array}$$

Multiplying this + adding remainder divided by denom gives a function equivalent to OG function

$6x - 9$ ← "remainder"

$$\overbrace{= \int (x-6) + \frac{6x-9}{x^2-1} dx}^{= \text{ via PLD}} = \underbrace{\int x+6 \, dx}_{\text{This can easily be integrated}} + \int \frac{6x-9}{x^2-1} dx$$

$$\int \frac{6x-9}{(x+1)(x-1)} dx = \int \frac{A}{x+1} dx + \int \frac{B}{x-1} dx = A\ln|x+1| + B\ln|x-1| + C$$

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Polynomial Long division